

ICHA, Russian Federation

WMO: 324110

Lat: 55.6137N

Lon: 155.6132E

Elev: 18

StdP: 14.69

Time Zone: 12.00 (E12)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-7.6	-17.0	2.2	-10.5	-13.2	2.7	-7.3	33.6	15.8	28.5	23.2	7.8	30	0.838

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	62.6	59.1	60.2	57.7	58.5	56.4	60.1	61.4	58.2	59.6	56.7	58.1	9.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.4	75.7	60.6	57.5	70.5	58.9	56.1	67.0	57.7	26.5	61.3	25.3	59.9	24.3	58.2	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.2	25.6	22.3	-15.8	69.4	4.1	4.5	-18.7	72.6	-21.1	75.2	-23.3	77.7	-26.3	80.9		
(5)			-15.6	64.7	3.9	3.9	-18.4	67.5	-20.7	69.8	-22.9	72.0	-25.8	74.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.7	10.7	13.1	18.6	28.7	37.2	45.7	52.7	54.0	48.2	39.4	27.0	15.9		
	DBStd	16.57	10.49	10.37	7.83	5.59	4.04	4.17	3.78	4.04	4.24	5.32	7.92	9.74		
	HDD50	6582	1217	1033	972	640	397	139	14	13	81	329	690	1057		
	HDD65	11788	1682	1453	1437	1090	861	579	383	342	505	794	1140	1522		
	CDD50	268	0	0	0	0	0	10	96	136	26	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	1	0	0	0	0	0	0	0	1	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	10.4	9.7	9.7	10.1	10.2	10.7	9.2	8.8	10.1	10.2	11.3	13.7	11.4		
	PrecAvg	28.4	1.1	0.8	0.9	1.5	1.7	2.0	3.5	4.0	3.3	3.9	3.3	2.0		
	PrecMax	41.7	2.7	4.1	2.8	3.9	4.4	5.2	10.3	9.1	8.2	8.2	8.8	5.7		
(15) Precipitation	PrecMin	17.0	0.1	0.0	0.1	0.2	0.2	0.0	0.2	0.4	0.6	1.1	0.9	0.2		
	PrecStd	6.4	0.7	0.8	0.7	0.9	1.1	1.3	2.2	2.2	1.9	1.6	1.8	1.4		
	DB	36.3	37.9	38.6	45.6	54.1	61.8	68.0	67.9	59.3	53.1	46.4	38.0			
	MCWB	31.7	32.7	33.4	39.9	46.2	53.1	63.0	63.8	55.4	46.7	43.7	32.7			
	2% DB	32.5	34.4	34.2	41.2	49.4	56.2	62.7	62.9	56.7	50.0	42.9	35.0			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	28.9	30.2	30.9	36.3	43.6	51.6	59.7	59.8	53.9	46.7	40.7	31.4			
	5% DB	28.7	31.1	31.5	38.7	46.4	53.3	60.1	60.7	55.5	48.2	39.4	31.4			
	MCWB	26.4	27.9	29.3	34.7	42.7	50.0	57.8	58.4	53.1	45.2	37.4	29.1			
	10% DB	25.2	27.8	29.2	36.3	43.7	51.3	58.2	59.2	54.3	46.7	37.2	28.8			
	MCWB	23.8	25.8	27.6	33.3	40.8	49.0	56.2	57.3	52.2	44.4	35.2	27.3			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	32.1	33.3	33.7	40.2	47.6	55.1	64.7	64.5	56.9	49.8	44.5	33.4			
	MCDB	35.5	37.5	37.5	44.8	52.7	59.6	67.0	65.6	58.0	51.0	45.7	36.5			
	2% WB	29.0	30.3	31.3	36.9	44.8	52.2	60.0	60.7	55.0	47.3	40.7	31.3			
	MCDB	31.8	33.8	33.7	40.3	48.4	54.9	61.5	62.0	55.9	48.8	42.6	34.1			
	5% WB	26.7	28.1	29.6	34.9	42.7	50.7	58.1	58.8	53.9	45.9	37.8	29.7			
(33) Mean Daily Temperature Range	MCDB	28.6	30.7	31.3	37.7	45.8	52.7	59.7	60.0	54.9	47.6	39.5	31.6			
	10% WB	23.7	25.8	27.8	33.6	40.9	49.3	56.5	57.5	52.7	44.6	35.3	27.4			
	MCDB	25.4	27.7	29.2	36.0	43.4	51.0	57.9	58.7	53.8	46.5	36.8	28.9			
	5% DB	11.2	11.8	10.9	10.7	12.8	10.9	9.7	9.0	9.3	9.6	9.2	10.1			
	MCWBR	9.9	9.3	8.9	7.4	8.2	7.2	7.1	6.3	7.0	7.5	8.0	8.6			
(40) Clear-Sky Solar Irradiance	5% WB	10.5	11.5	10.2	9.4	11.2	9.7	8.8	7.6	7.2	7.4	8.6	9.6			
	MCWBR	9.6	9.5	8.7	6.9	7.9	6.9	7.0	6.0	6.3	6.5	8.0	8.4			
	taub	0.272	0.311	0.357	0.433	0.476	0.445	0.460	0.432	0.367	0.363	0.327	0.278			
	taud	2.148	2.092	2.028	1.911	1.948	2.141	2.142	2.225	2.383	2.209	2.095	2.082			
(42) All-Sky Solar Radiation	Ebn at Noon	200	233	249	245	243	254	247	246	250	212	168	161			
	Edh at Noon	19	30	40	53	54	45	44	38	29	26	20	16			
	RadAvg	191	416	852	1225	1502	1537	1413	1078	870	467	202	126			
(45)	RadStd	22	57	100	143	188	135	147	115	75	43	23	13			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page